Lab - AzCopy Tool
To Begin with the lab

Summary
In this lab, you learn how to use the **AzCopy** command-line utility to transfer data between a local machine and an Azure Storage Account. You install the AzCopy tool, create a Blob Container, and upload a file to Azure Blob Storage using a **Shared Access Signature (SAS)** token. AzCopy provides a fast and efficient way to manage Azure Storage resources from the command line.

Understanding
AzCopy is a Microsoft command-line tool used to copy files and folders between local storage and Azure Storage services such as **Blob Storage** and **Azure Files**. It is available for **Windows**, **Linux**, and **macOS**.
Unlike Azure Storage Explorer, which provides a graphical interface, AzCopy is designed for developers and administrators who need to automate storage operations or transfer large amounts of data efficiently.
AzCopy uses a **Shared Access Signature (SAS)** token to authenticate requests without exposing the Storage Account access keys. With AzCopy, you can perform operations such as creating containers, uploading blobs, downloading blobs, synchronizing data, and copying files between storage accounts.

Example
Suppose you have a local SQL script named **init.sql** that needs to be uploaded to Azure Blob Storage.
Local File:
D:\Dev3\init.sql
Storage Account:
stdeveastus10
Blob Container:
scripts
Blob:
init.sql
Application Flow:

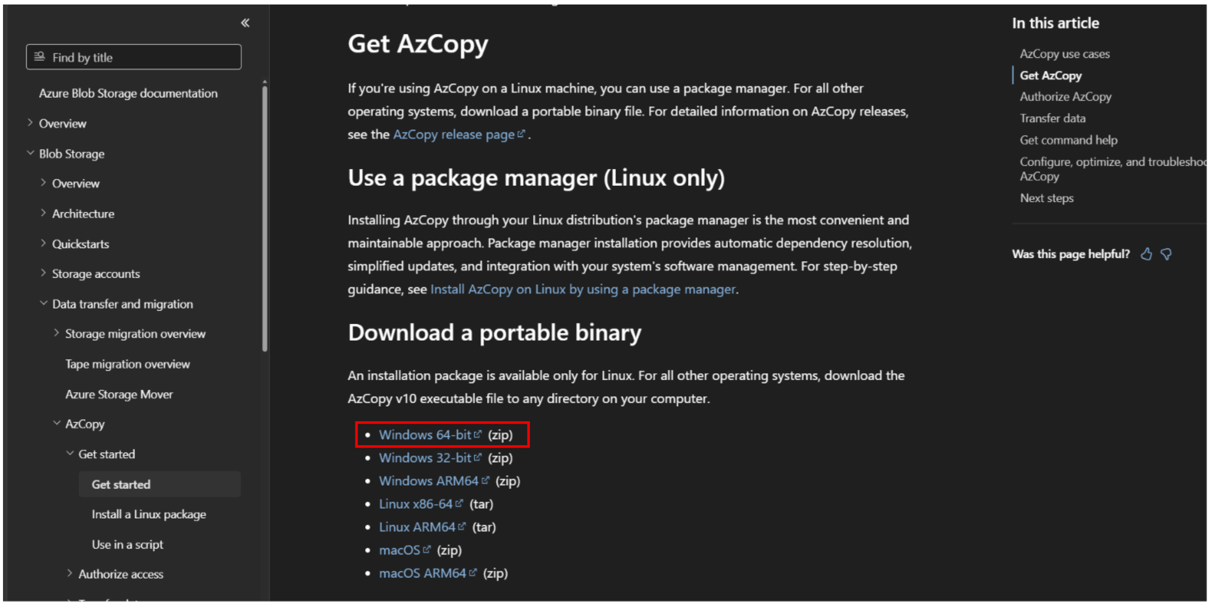
```
Local Machine
      |
      ▼
AzCopy Tool
      |
      ▼
Azure Storage Account
      |
      ▼
Blob Container (scripts)
```

|
▼
init.sql (Blob)
The **init.sql** file is uploaded to the **scripts** container and becomes available in Azure Blob Storage.

Lab Steps

Step 1: Download AzCopy

- Open the AzCopy download page.
- Download the appropriate version for your operating system.
- For Windows, download the **Windows 64-bit ZIP** package.



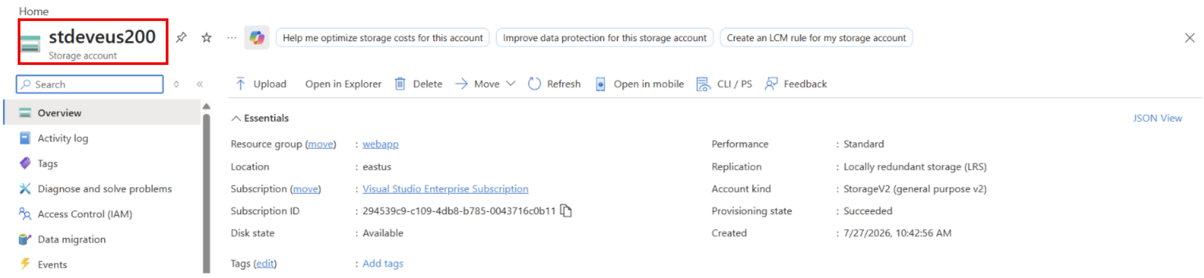
- Extract the ZIP file.
- Copy **AzCopy.exe** to a convenient folder (for example, **D:\Dev3**).

Step 2: Prepare the Local File

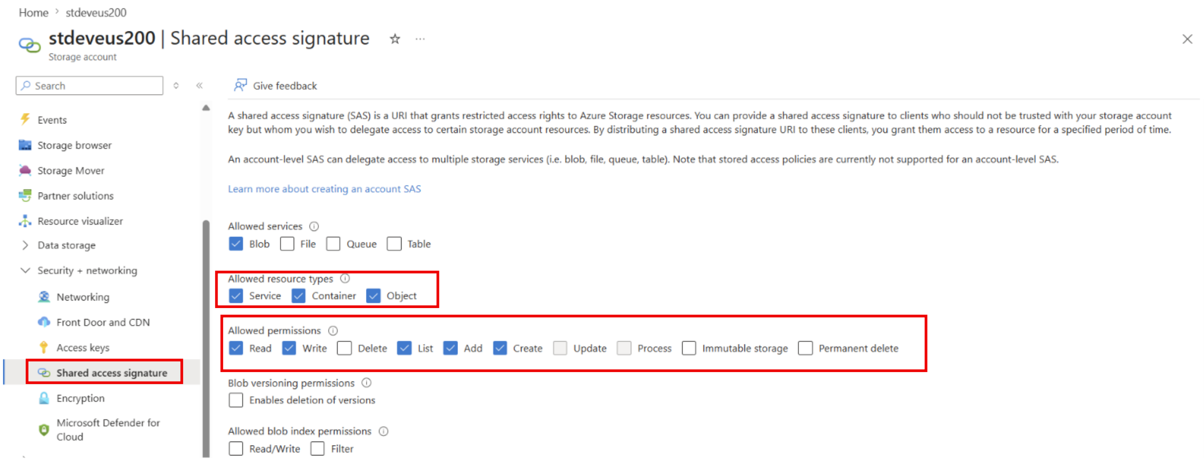
- Place the file you want to upload in the same directory as **AzCopy.exe**.

Step 3: Generate a SAS Token

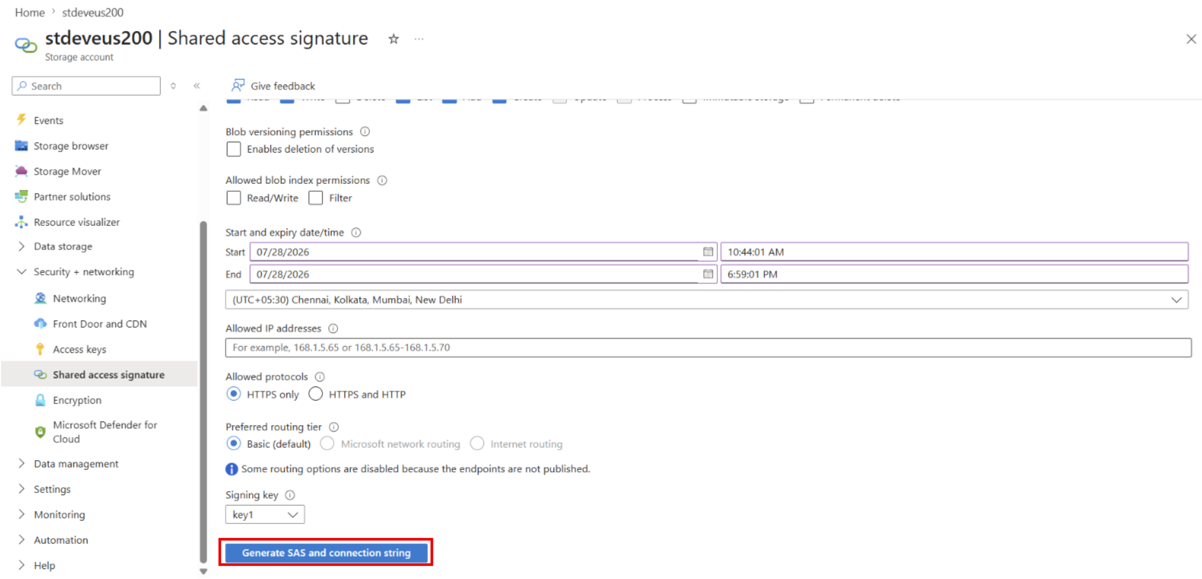
- Open the **Azure Portal**.
- Navigate to your **Storage Account**.



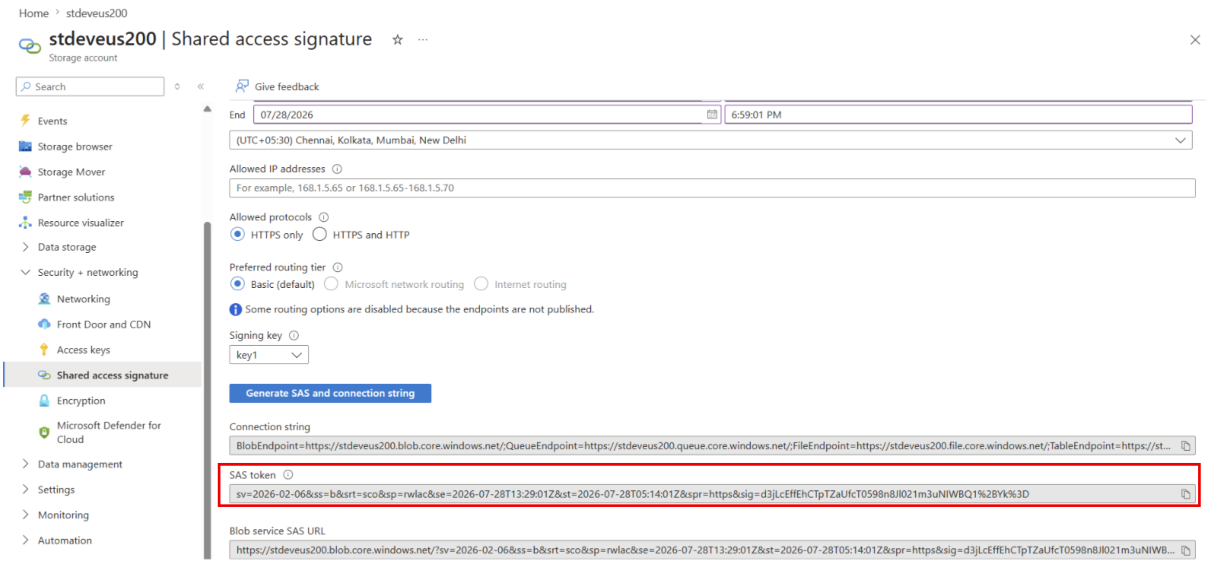
- Select **Shared access signature**.
- Select **Blob** as the allowed service.
- Select the required **resource types**.
- Choose the **required permissions** (**Read, Write, Create, List, etc**.).



- Click **Generate SAS and connection string**.



- Copy the generated **SAS Token**.



Step 4: Create a Blob Container Using AzCopy

Run the following command:

azcopy make "https://<storage-account>.blob.core.windows.net/scripts?<SAS-Token>"

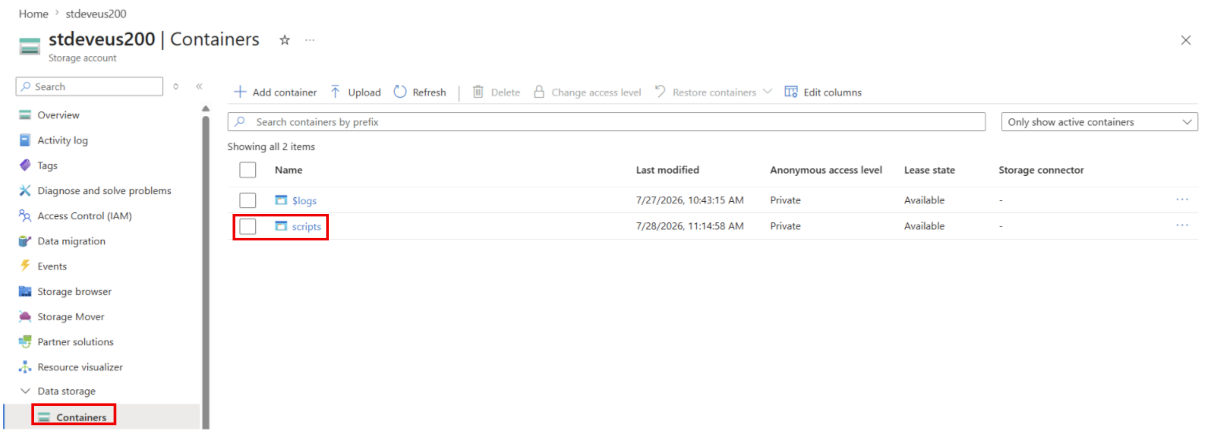
Example:

azcopy make "https://**stdeveus200**.blob.core.windows.net/scripts? sv=2026-02-06&ss=b&srt=sco&sp=rwlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D"

- Creates a new Blob Container.
- Uses the SAS Token for authentication.

Step 5: Verify the Container

- Open the Azure Portal.
- Navigate to **Storage Account → Containers**.
- Refresh the page.
- Verify that the **scripts** container has been created.



Step 6: Upload a Blob Using AzCopy

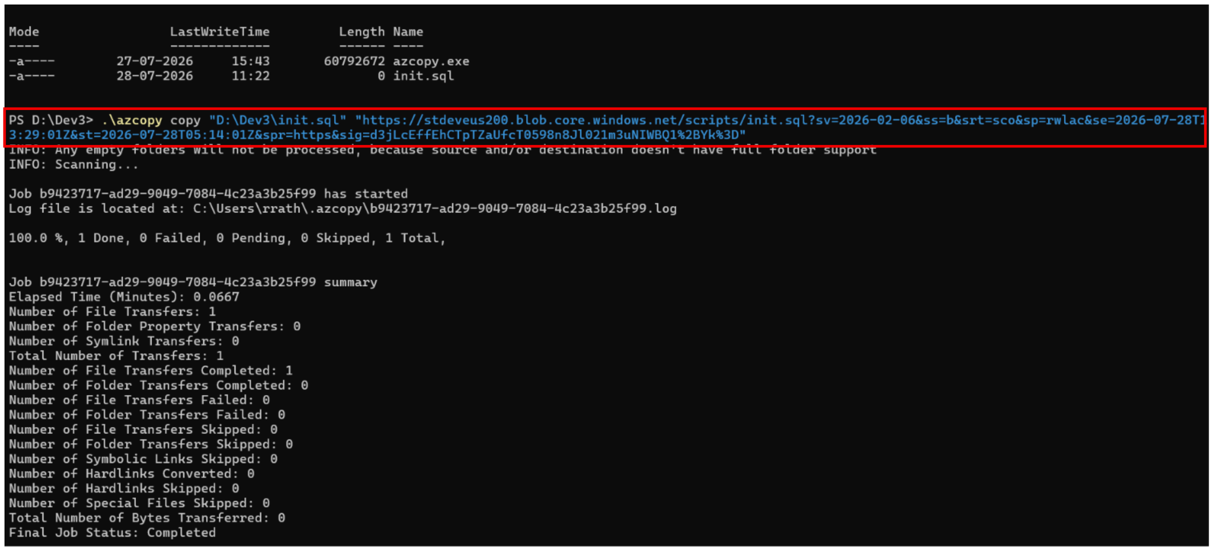
Run the following command:

azcopy copy " " "https://<storage-account>.blob.core.windows.net/scripts/init.sql?<SAS-Token>"

Example:

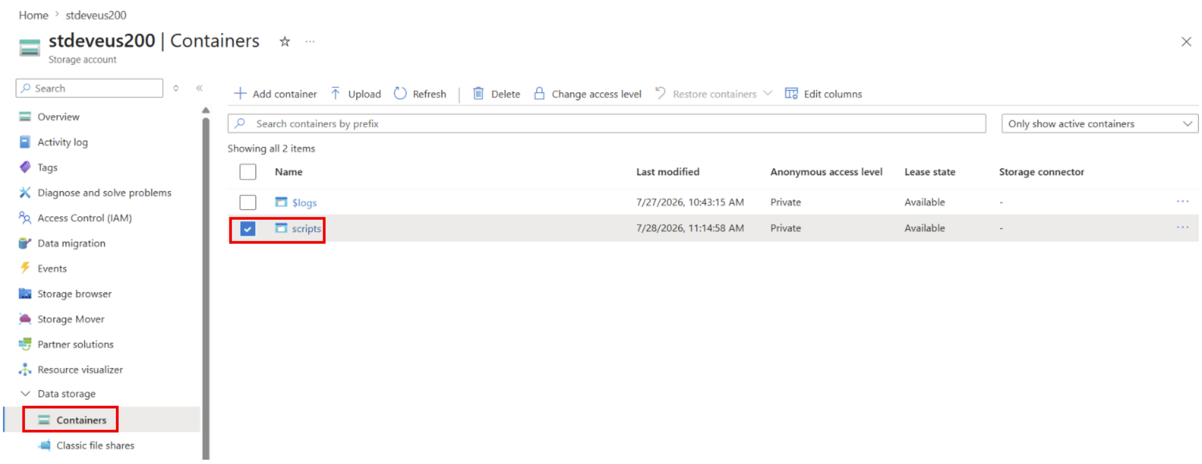
azcopy copy "D:\Dev3\init.sql" "https://stdeveus200.blob.core.windows.net/scripts/init.sqlsv=2026-02-06&ss=b&srt=sco&sp=rwlac&se=2026-07-28T13:29:01Z&st=2026-07-28T05:14:01Z&spr=https&sig=d3jLcEffEhCTpTZaUfcT0598n8Jl021m3uNIWBQ1%2BYk%3D"

- Uploads the local file.
- Stores it in the **scripts** Blob Container.

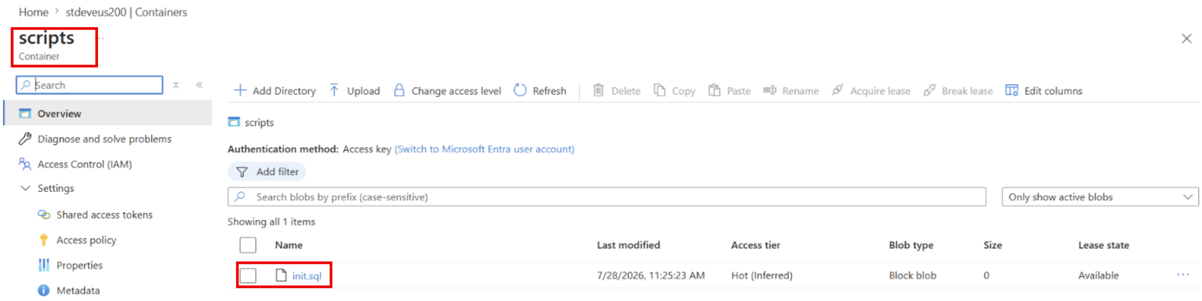


Step 7: Verify the Uploaded Blob

- Open the Azure Portal.
- Navigate to **Storage Account → Containers → scripts**.
- Refresh the page.



- Verify that **init.sql** appears in the container.


Home > stdeveus200 | Containers
scripts
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Shared access tokens
Access policy
Properties
Metadata
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Upload
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Delete
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Acquire lease
Break lease
Edit columns
scripts
Authentication method: Access key (Switch to Microsoft Entra user account)
Add filter
Search blobs by prefix (case-sensitive)
Only show active blobs
Showing all 1 items
Name
Last modified
Access tier
Blob type
Size
Lease state
init.sql
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Block blob
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Available